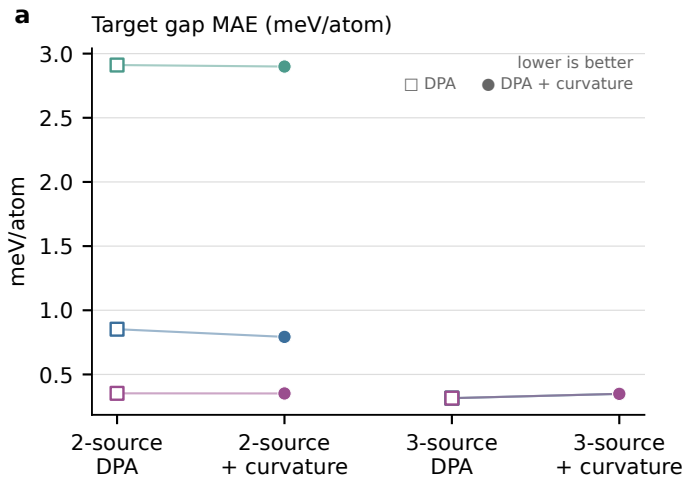
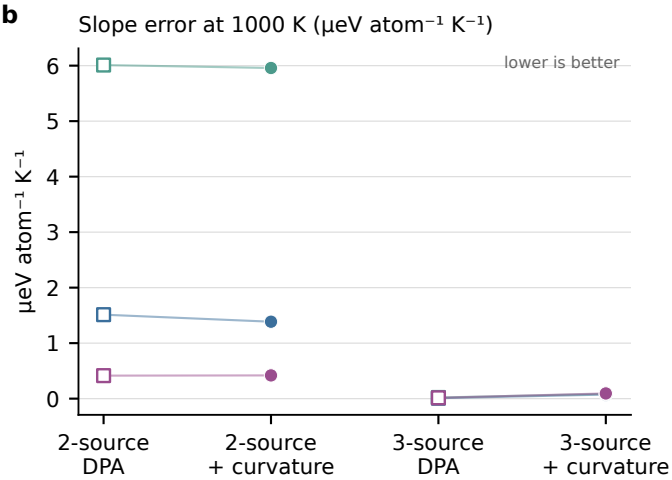
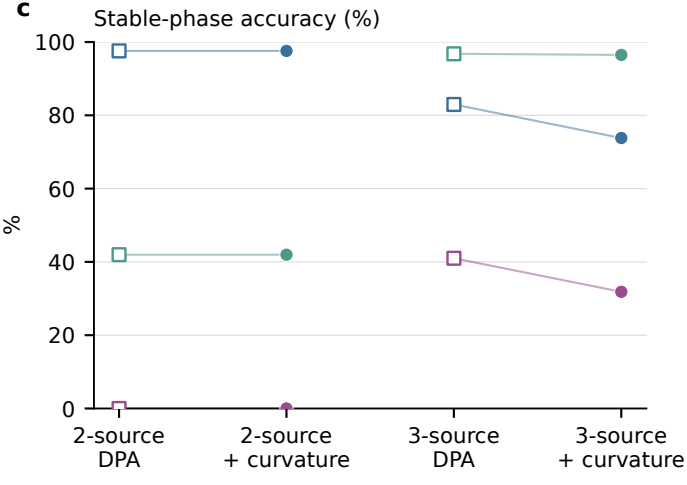
Frozen curvature queries add small gap gains in the two-source fit, but do not improve phase decisions







d Identity- and direction-aware events

Source pair	2src DPA	2src +Q	3src DPA	3src +Q
quartz + $P6_3$	1/0/0	1/0/0	0/1/2	0/1/2
quartz + C2221	0/1/1	0/1/1	1/0/0	1/0/0
$P6_3$ + C2221	0/0/0	0/0/0	0/0/1	0/0/1

Cells: hits / missed / false; identity + direction, 100 K match window.